Solutions and teaching notes for chapter 9

Teacher use. The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. $2^7 = 128$. *Action:* Rebuild the product of seven factors.
2. $\frac{1}{9}$. *Action:* Go back to the reciprocal power.
3. $x = 2$. *Action:* Isolate the unknown while preserving equivalence.
4. $2x + 2$. *Action:* Apply the sum of the derivatives.
5. 0.8. *Action:* Subtract the rate from 1.
6. 5×2^n. *Action:* Count the n multiplications.

Investigation activity

The condition $f\prime = f$ motivates $\Delta y \approx hy$. With h=$\frac{1}{4}$, the values are 1; 1.25; 1.5625; 1.953125. This construction approximates the function and does not replace its characterisation.

Immediate applications

1. $y = e + e(x - 1) = ex$.
2. e^{2x+1}; $x + 2 > 2x$, so $x < 2$.
3. $g\prime(t) = -6e^{-2t} < 0$ on $\mathbb{R}$: strictly decreasing.
4. $u_n = 5(e^{0.3})^n$; the ratio over two units is $e^{0.6}$.
5. Apply the inequality proved with the argument $2x$; equality only at x=0.

Exercises: method and results

Exercise 9.1 – Transforming. e^2; e^{5x}; $1 + e^{-x}$. For the last expression, distribute then simplify the products.

Exercise 9.2 – Equations and inequalities. By strict increase: $2x - 1 = x + 3$, so x=4. Then $-3x \geq x - 2$, that is $x \leq \frac{1}{2}$.

Exercise 9.3 – Product and maximum. $f\prime = (1 - x)e^{-x}$; increasing up to 1 then decreasing. Maximum $f(1) = \frac{1}{e}$. Values at the endpoints 0 and $4e^{-4}$ are smaller.

Exercise 9.4 – A modelled population. 200; factor $e^{0.1}$; rate $e^{0.1} - 1 \approx 10.52\,\%$. Resources and the carrying capacity of the environment are not taken into account; this is a model over a limited time. Both curves requested pass through (0;1); the first increases, the second decreases. They are symmetric about the vertical axis; see the figure in the section From discrete to continuous.

Exercise 9.5 – Searching for a threshold. Test n=0,1,... in turn with an iteration limit. $Q(4) \approx 10.8268 \geq 10$, $Q(5) \approx 6.5668 < 10$; decrease proved, so the first integer is 5.

Exercise 9.6 – A symmetry. $f(-x) = f(x)$: even. $f\prime = e^x - e^{-x}$ has the sign of x, so minimum 2 at 0. Another proof: $f(x) - 2 = (e^{\frac{x}{2}} - e^{-\frac{x}{2}})^2$.

Exercise 9.7 – Uniqueness under a hypothesis. $g\prime = f\prime e^{-x} - fe^{-x} = 0$. On $\mathbb{R}$, g is constant, and g(0)=1. Thus $f(x) = e^x$. This proof of uniqueness uses an exponential already constructed; it does not prove existence.

Exercise 9.8 – Delayed review. $\frac{u_{n+1}}{u_n} = e^2$; first term e. Sum $e(1 + e^2 + e^4) = \frac{e(1 - e^6)}{1 - e^2}$. The direct sum is just as efficient here.

CQFD, exercise 18 p. 187, questions a, b and c. a: e^{3x}. b: e^2, because e^{x+2} is divided by e^x, which is never zero. c: e^{x+1}. Do not confuse the number e with the variable x.

CQFD, exercise 30 p. 187, questions a, b and c. a: $x \leq 0$. b: $2x > 1$, so $x > \frac{1}{2}$. c: $-3x \geq 2x + 1$, so $x \leq -\frac{1}{5}$. The strict increase of the exponential preserves the direction; dividing by -5 reverses it in c.

MCQ: answers and reasons

MCQ 1.

1. B. Initial condition.
2. A. Addition property.
3. D. Factor -2.
4. C. Strict increase and $e^0 = 1$.
5. B. Positivity of the exponential.

MCQ 2.

1. C. Compare the exponents.
2. A. Ratio of consecutive terms.
3. D. Value 1 and slope 1.
4. B. Product rule.
5. C. Minimum of the difference.

True or false

1. False: at 0, 1 is different from 2.
2. True: its derivative $-3e^{-3t}$ is negative.
3. False: the discrete rate is $e^{0.1} - 1$, about 10.52 %.

Differentiated teaching. For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.